

THE SCOTS COLLEGE



Assessment Task

Year 12

Ext-1 Mathematics

HSC Task 1: Tuesday 21st February,
Week 4, Term 1 2012

Weighting of Task:10%

Time allowed: 45 minutes

Directions:

- All 12 questions must be attempted on your own paper.
- Questions 1 to 5 are Multiple Choice A, B, C, or D only.
- Write your name on each page.
- Start a new page for questions 6 to 12.
- Show all necessary working in questions 6 to 12.
- Approved calculators may be used.

Topics Assessed

Integration of Trigonometric functions.(Q,9)	/5
Solution of trigonometric equations.(Q,3,6,12)	/8
Polynomials and Numerical Estimation of Roots.(Q,1,7,10)	/10
Inverse Functions and Inverse Trigonometric Functions.(Q,2,4,5,8,11)	/17

Total Marks:**/40**

Questions 1 to 5 ONLY are multiple choice (1 mark each).
For questions 1 to 5, write A, B, C or D on your answer sheet.

1. When $x^2 + 4x + 5b$ is divided by x the remainder is -10 .

The value of b is

- (A) 2 (B) -12 (C) -2 (D) 0

2. The domain of $y = -2\sin^{-1}(1 - 4x)$ is

- (A) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (B) $-\frac{1}{2} \leq x \leq \frac{1}{2}$ (C) $0 \leq x \leq \frac{1}{2}$ (D) $-\frac{\pi}{8} \leq x \leq \frac{\pi}{8}$

3. $2\cos(t - \frac{\pi}{3})$ can be rewritten as

- (A) $\cos t - \sqrt{3}\sin t$ (B) $2\cos t - 2\cos\frac{\pi}{3}$
 (C) $\cos t + \sqrt{3}\sin t$ (D) $\sin t - \sqrt{3}\cos t$

4. $\cos^{-1}(-x) =$

- (A) $\pi - \cos^{-1}x$ (B) $\pi + \cos^{-1}x$ (C) $\sin^{-1}x$ (D) $-\sin^{-1}x$

5. $\sin(2\tan^{-1}\frac{2}{3}) =$

- (A) $\frac{\sqrt{13}}{12}$ (B) $\frac{12}{\sqrt{13}}$ (C) $\frac{6}{\sqrt{13}}$ (D) $\frac{12}{13}$

End of Multiple Choice Section

For Questions 6 to 12 start each question on a new page and show all working.

6.

$$5\cos\theta + 12\sin\theta = R\cos(\theta - \alpha) \text{ where } R > 0 \text{ and } \alpha \text{ is acute.}$$

(a) Find the value of R . (2)

(b) Find the size of α to 3 decimal places. (1)

(c) Hence solve to 3 decimal places

$$5\cos\theta + 12\sin\theta = 13 \text{ for } 0 \leq \theta \leq 4\pi. \quad (2)$$

7.

(a) Show that $\sin x = x - 1$ has a root near $x = 2$ (2)

(b) Use Newton's Method once to find a better approximation to this root.
(Answer to 2 decimal places) (2)

8.

The function $f(x) = 3x - x^3$ has a minimum turning point at $(-1, -2)$ and a maximum turning point at $(1, 2)$.

(a) Sketch $y = f(x)$ showing given turning points and intercepts on both axes. (2)

(b) Find the largest domain containing the origin for which $f(x)$ has an inverse function $y = f^{-1}(x)$. (1)

(c) Find the domain and range of $y = f^{-1}(x)$. (2)

(d) Sketch $y = f^{-1}(x)$ clearly showing the end points. (3)

9.

(a) Find $\int \sin^2 3x \, dx$ (2)

(b) Find in terms of π the volume of the solid formed when $y = \cos x$ is rotated about the x-axis from $x = 0$ to $x = \frac{\pi}{2}$. (3)

10.

(a) If α, β, γ are the roots of $x^3 + 4x - 9 = 0$ find $\alpha(\beta+1) + \beta(\gamma+1) + \gamma(\alpha+1)$ (2)

(b) Given that $Q(x) = 4x^3 + kx + 6$ has a root at $x = -3$.
(i) Find k . (1)

(ii) Write $Q(x)$ in the form $(x+3)(ax^2 + bx + c)$ (2)

11. Given the function $f(x) = 2 \sin^{-1}\left(\frac{x}{3}\right)$

(a) Find $f(0)$ (1)

(b) State the domain and range of $y = f(x)$ (2)

(c) Draw the graph of $y = f(x)$ showing the end points. (3)

12. Given $\sqrt{2} \cos \theta = 1$

(a) Write the general solution for this equation in terms of π . (1)

(b) Solve for $n = -1$ and $n = 2$. Answer in terms of π . (1)

END OF EXAMINATION

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1 $p(x) = \frac{2012}{-10} \text{ Ext-1 HSC Task 1 - Solution}$

$5b = -10$
 $b = -2$

(C)

Q2 $-1 \leq 1 - 4x \leq 1$

$-2 \leq -4x \leq 0$

$\frac{1}{2} \geq x \geq 0$

(C)

Q3 $2 \cos\left(t - \frac{\pi}{3}\right) = 2 \cos t \cos \frac{\pi}{3} + 2 \sin t \sin \frac{\pi}{3}$
 $= 2 \cos t \times \frac{1}{2} + 2 \sin t \times \frac{\sqrt{3}}{2}$
 $= \cos t + \sqrt{3} \sin t$

(C)

Q4 $\cos^{-1}(-x) = \pi - \cos^{-1}x$

(A)

Q5 $\sin\left(2 \tan^{-1} \frac{2}{3}\right) = \sin 2B$

where $\tan B = \frac{2}{3}$

$\therefore \cos B = \frac{3}{\sqrt{13}}$

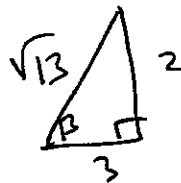
$\sin B = \frac{2}{\sqrt{13}}$

$= 2 \sin B \cos B$

$= 2 \times \frac{2}{\sqrt{13}} \times \frac{3}{\sqrt{13}}$

$= \frac{12}{13}$

(D)



Q6 (a) $\frac{5}{R} \cos \theta + \frac{12}{R} \sin \theta = \cos(\theta - \alpha)$ (1)

$= \cos \theta \cos \alpha + \sin \theta \sin \alpha$

$\therefore \cos \alpha = \frac{5}{R} \quad \sin \alpha = \frac{12}{R}$

$\therefore R = \sqrt{12^2 + 5^2}$ (1)
 $R = 13$

(b) $\tan \alpha = \frac{12}{5}$ (1)
 $\alpha = 1.176$



(c) (1)

$\therefore 13 \cos(\theta - 1.176) = 1$

$\cos(\theta - 1.176) = \frac{1}{13}$

$\theta - 1.176 = 0, 2\pi$

$\theta = 1.176, 2\pi + 1.176$

$\theta = 1.176^\circ, 7.459^\circ$

7. (a) $f(x) = \sin x - x + 1$

$f(1.9) = 0.046$ above x -axis (1)

$f(2.1) = -0.0237$ below x -axis (1)

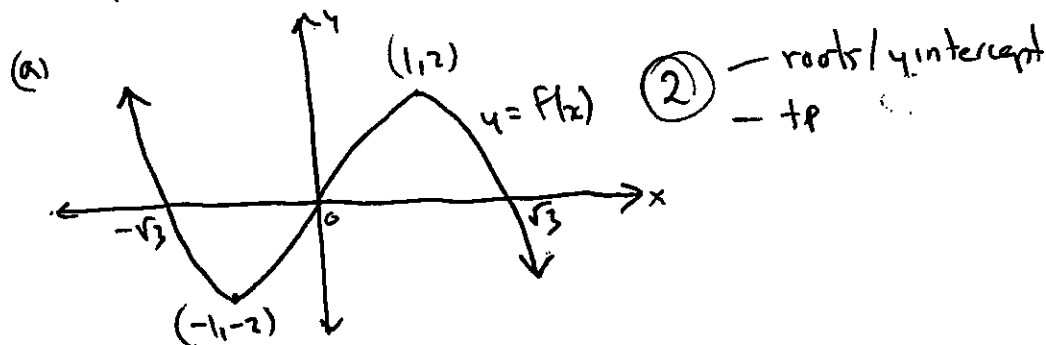
$\therefore$ root exists between 1.9 and 2.1.

(b) $a_1 = a - \frac{f(a)}{f'(a)}$

$= \frac{2 - (\sin 2 - 2 + 1)}{\cos 2 - 1}$ (1)

$= 1.94$ (1)

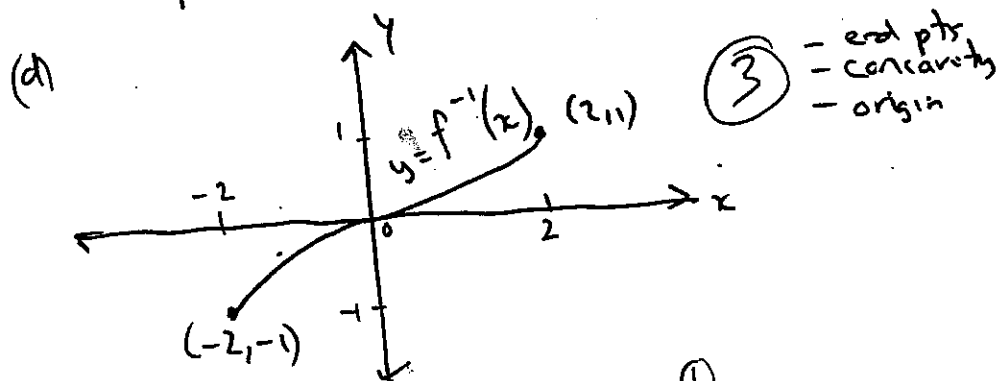
8. $f(x) = 3x - x^3$



(b) $D_f : -1 \leq x \leq 1$ (1)

(c) $D_{f^{-1}} : -2 \leq x \leq 2$ (2)

$R_{f^{-1}} : -1 \leq y \leq 1$



9. (a) $\int \sin^2 3x \, dx = \frac{1}{2} \int (1 - \cos 6x) \, dx$ (1)

$= \frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + c$

$= \frac{1}{2} x - \frac{1}{12} \sin 6x + c$ (1)

7. (a) $f(x) = \sin x - x + 1$

$f(1.9) = 0.046$ above x -axis (1)

$f(2.1) = -0.0237$ below x -axis (1)

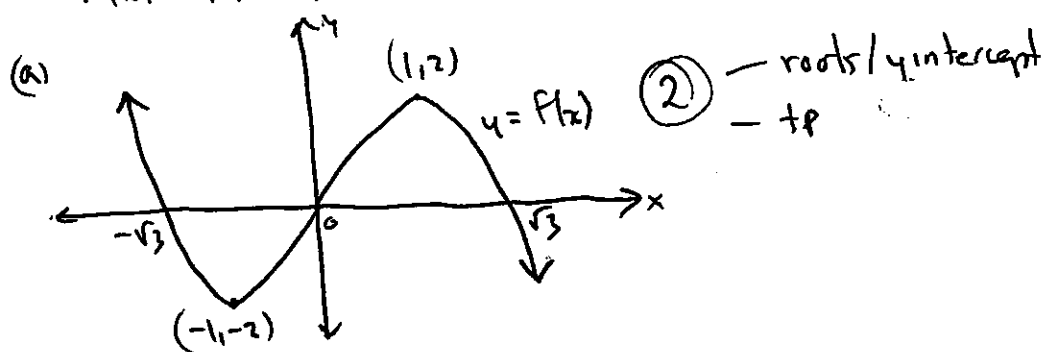
$\therefore$ root exists between 1.9 and 2.1.

(b) $a_1 = a - \frac{f(a)}{f'(a)}$

$= 2 - \frac{(\sin 2 - 2 + 1)}{\cos 2 - 1}$ (1)

$= 1.94$ (1)

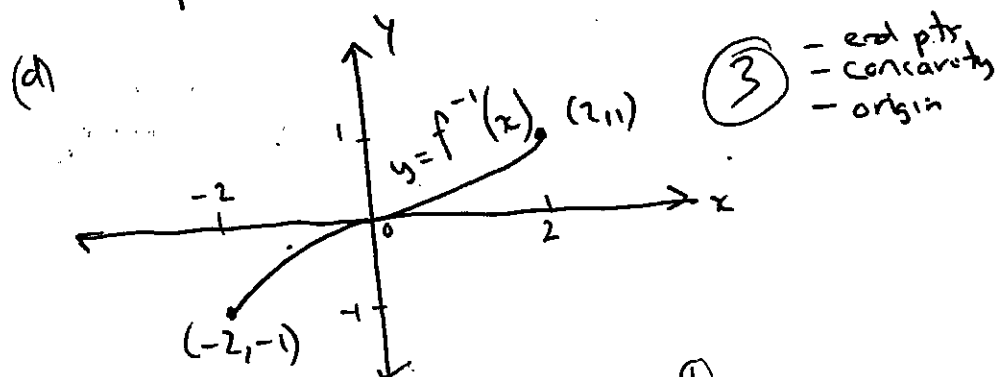
8. $f(x) = 3x - x^3$



(b) $D_f: -1 \leq x \leq 1$ (1)

(c) $D_{f^{-1}}: -2 \leq x \leq 2$ (2)

$R_{f^{-1}}: -1 \leq y \leq 1$

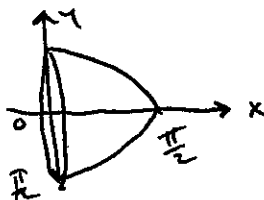


9. (a) $\int \sin^2 3x \, dx = \frac{1}{2} \int (1 - \cos 6x) \, dx$ (1)

$= \frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + c$

$= \frac{1}{2} x - \frac{1}{12} \sin 6x + c$ (1)

9. (b)



$$\begin{aligned}
 V &= \pi \int_0^{\pi/2} \cos^2 x \, dx \\
 &= \frac{1}{2} \pi \int_0^{\pi/2} (1 + \cos 2x) \, dx \\
 &= \frac{1}{2} \pi \left[x + \frac{1}{2} \sin 2x \right]_0^{\pi/2} \\
 &= \frac{1}{2} \pi \left[\left(\frac{\pi}{2} + \frac{1}{2} \sin \pi \right) - \left(0 + \frac{1}{2} \sin 0 \right) \right] \\
 &= \frac{1}{2} \pi \left[\frac{\pi}{2} + \frac{1}{2} \right] \\
 V &= \frac{\pi^2 + \pi}{4} \text{ u}^3.
 \end{aligned}$$

10. (a) $\alpha(\beta+1) + \beta(\gamma+1) + \gamma(\alpha+1) = \alpha\beta + \alpha + \beta\gamma + \beta + \gamma\alpha + \gamma$
 $= \alpha\beta + \beta\gamma + \gamma\alpha + \alpha + \beta + \gamma$
 $= 4 + 0$
 $= 4.$

(b) (i) $0 = -108 - 3k + 6$
 $3k = -102$
 $k = -34.$

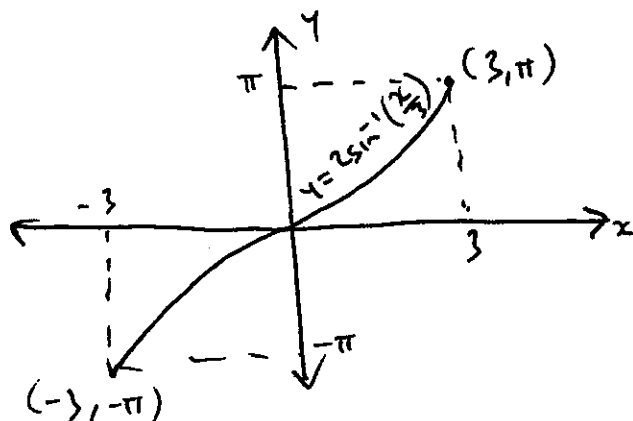
(iii)
$$\begin{array}{r}
 4x^2 - 12x + 2 \\
 x+3 \overline{) 4x^2 - 34x + 6} \\
 \underline{4x^2 + 12x + 6} \\
 -46x + 0
 \end{array}$$
 $\therefore Q(x) = (x+3)(4x^2 - 12x + 2)$

11. (a) $f(0) = 2\sin^{-1}(0)$
 $f(0) = 0$

(b) $D_f: -1 \leq \frac{x}{3} \leq 1$
 $-3 \leq x \leq 3$

$R_f: -\frac{\pi}{2} \leq \frac{y}{2} \leq \frac{\pi}{2}$
 $-\pi \leq y \leq \pi.$

(c)



3 - endpoints
 - origin
 - continuity.

12 $\cos \theta = \frac{1}{\sqrt{2}}$
 (a) $\theta = 2n\pi \pm \cos^{-1}(\frac{1}{\sqrt{2}})$
 $\theta = 2n\pi \pm \frac{\pi}{4}$

(b) $n = -1 \quad n = 2$
 $\theta = -2\pi \pm \frac{\pi}{4}, \theta = 4\pi \pm \frac{\pi}{4}.$